ASSIGNMENT MANUAL

Digital Image Processing (20ECE304T)

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Submitted by

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Submitted to

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EDGE ENHANCEMENT
(Using Sobel and Laplacian Filters)

1 AIM

To apply Laplacian and Sobel filters on smooth grayscale images to detect and enhance the edges, and to compare the original and enhanced outputs for better understanding of image details.

2 METHODOLOGIES

2.1 Sobel Filter

Sobel filter, also known as Sobel operator, is used in image processing primarily for edge detection. These emphasize regions where the intensity changes sharply which helps in identifying object boundaries and structural features in images. This allows subsequent algorithms to:

- Recognize object boundaries and structure.
- Improve image analysis, segmentation, and feature extraction.
- Distinguish separate regions, aiding in automation and computer vision tasks.

2.1.1 How Sobel filters work

1. <u>Convolution with Kernels</u>
    - The Sobel filter works by convolving an image with two specific 3×3 kernels: one that detects horizontal changes (x-direction), and one for vertical changes (y-direction). The kernels are:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

    - Applying these to a grayscale image estimates the gradient (first derivative) of the image intensity at each pixel, resulting in two new images (G_x and G_y) which show how much the image intensity changes horizontally and vertically.

2. <u>Sliding Over Image (Convolution)</u>
    - For each pixel, take the 3×3 region around it, multiply each value by the corresponding value in the kernel, and add up the results. Do this for both kernels.
    - This gives two numbers: one for the horizontal gradient (G_x) and one for the vertical gradient (G_y).

3. Combine Gradients
   - At every pixel, combine these using:
     - Gradient magnitude (edge strength):

$$M(x, y) = \sqrt{(G_x[x, y]^2 + G_y[x, y]^2)}$$

     - Sometimes, for speed, the simpler approximation $|G_x| + |G_y|$ is used for the magnitude.

4. Result: Edge Map
   - High values of the combined gradient indicate pixels on or near edges. The output is a new image (the edge map), where brighter pixels represent stronger edges.

2.1.2 Applications

- Edge Detection**:** Highlights object boundaries, essential for segmentation and feature extraction tasks.
- Image Analysis**:** Used in focus and clarity assessment, especially in medical and quality assessment images.
- Computer Vision**:** Helps algorithms detect shapes, letters, and other patterns for further processing.
- Feature Extraction**:** Integrates with methods like Histogram of Oriented Gradients (HOG) for object recognition.

2.2 Laplacian Filter

A Laplacian filter is a digital image processing tool used primarily for edge detection by measuring how rapidly pixel intensity values change across an image. It operates as a second-order derivative filter, detecting changes in intensity in all directions and highlighting regions of rapid transitions—typically the edges.

2.2.1 How Laplacian filters work

- The Laplacian operator computes the sum of second derivatives in both horizontal and vertical directions.
- This is usually implemented using a convolution kernel like:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

  or its sign-reversed counterpart, such that the central value is positive and surrounded by negative ones or vice versa.

- When applied to an image, the filter calculates the rate at which adjacent pixel intensity values change, enabling it to locate edges in a single operation by emphasizing areas of rapid change.

2.2.2 Applications

- Edge Detection**:** Detecting edges by highlighting regions of rapid intensity change in the image.
- Feature Extraction**:** Integrates with methods like Histogram of Oriented Gradients (HOG) for object recognition.
- Image Sharpening: Enhancing the details and fine structures by emphasizing edges.
- Medical Imaging: Enhancing the visibility of structures in medical images like MRI or CT scans.

2.3 Edge Enhancement

The filters, Sobel and Laplacian are used for edge detection here. But our main aim is edge enhancement and not edge detection. After successfully detecting edges using these filters, we then add these (edge detected images) to the original images to enhance the edges. The formula for edge enhancement is given by:

$$\text{Enhanced image} = \text{Original image} + \alpha * \text{Edge detected image}$$

where α is the scaling factor which controls the intensity of enhancement and keeps the contents of the image visible.

3 CODE

4 RESULTS

Figure 1 shows the first original image, the edge detected image using Sobel filter and finally the edge enhanced image of the first image which was obtained by adding the edge detected image to the original image.

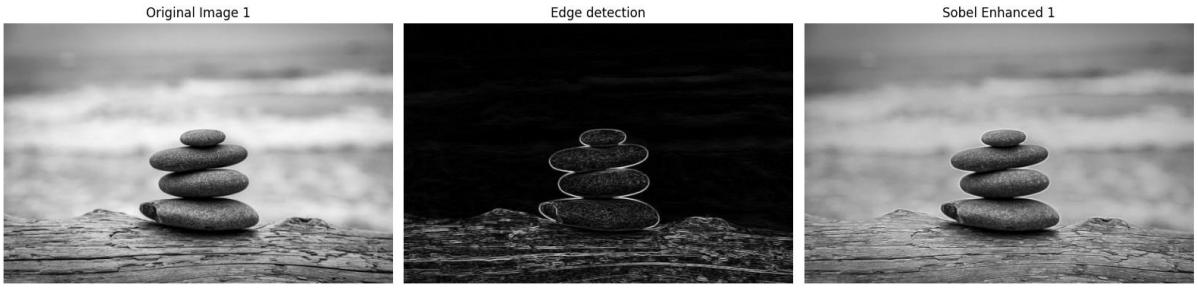

Figure 1: Stages of Edge Enhancement Using the Sobel Filter for Image 1

Figure 2 shows the second original image, the edge detected image using Sobel filter and finally the edge enhanced image of the second image which was obtained by adding the edge detected image to the original image.

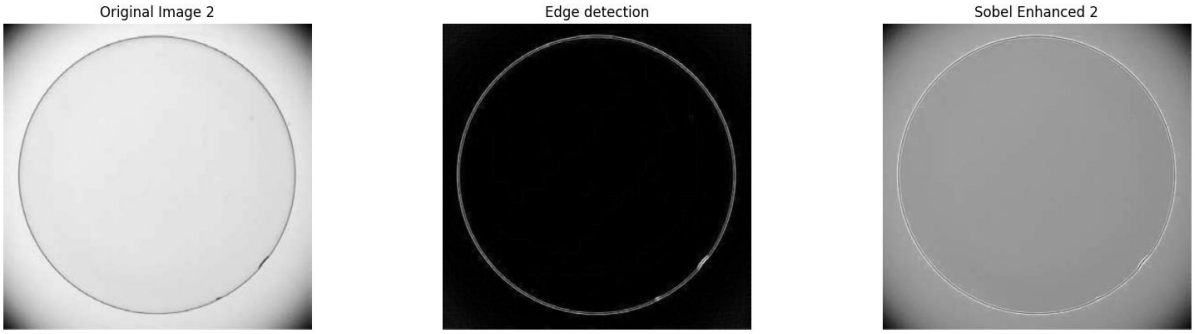

Figure 2: Stages of Edge Enhancement Using the Sobel Filter for Image 2

Figure 3 shows the first original image, the edge detected image using Laplacian filter and finally the edge enhanced image of the first image which was obtained by adding the edge detected image to the original image.

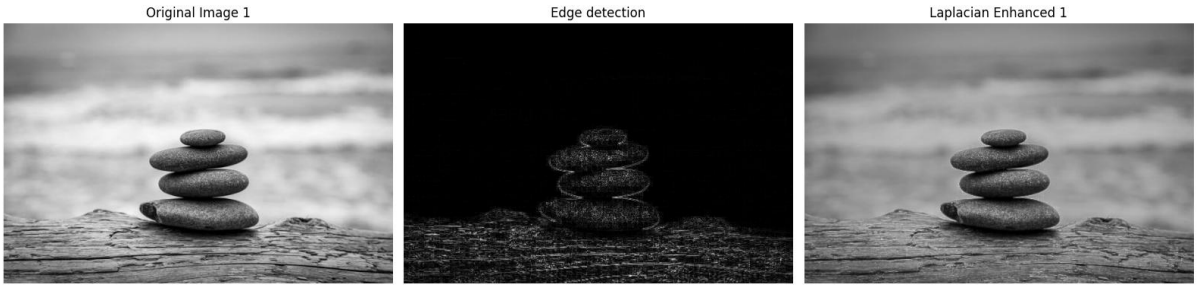

Figure 3: Stages of Edge Enhancement Using the Laplacian Filter for Image 1

Figure 4 shows the second original image, the edge detected image using Laplacian filter and finally the edge enhanced image of the second image which was obtained by adding the edge detected image to the original image.

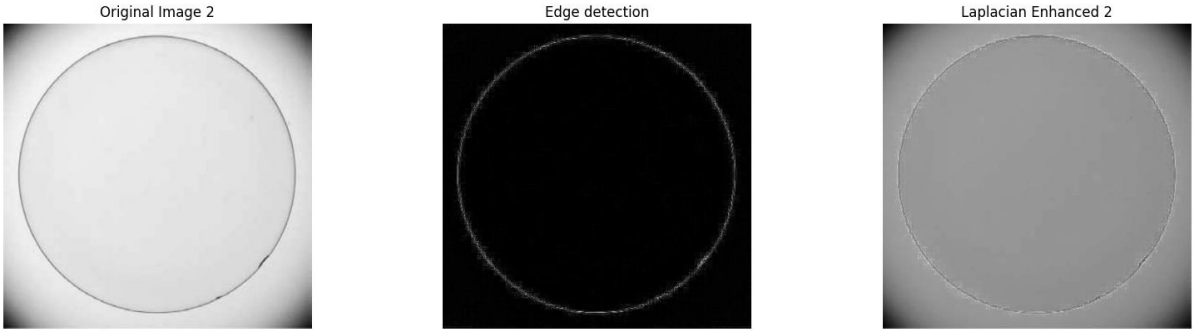

Figure 4: Stages of Edge Enhancement Using the Laplacian Filter for Image 2

5 DISCUSSIONS

As observed from the results, the enhanced edges in the second image are more visible than the first image. This is because the second image (original) is a nearly uniform circle on a plain background and has very little texture or detail except at the boundary of the circle. The filters strongly highlight this boundary with high gradient values, resulting in a much more pronounced effect. The relative difference between the edge and the background is high.

On the other hand, the first image (original) has a lot of natural texture, gradients and noise in both the foreground and background (wood, waves, sky). The filters detect edges everywhere, some are weak, some are strong, and many blend into the textured regions. In images with lots of natural texture, subtle edges may not be visually dominant when combined, as the enhancement competes with the underlying variability.

The edge detected images highlight only the edges of the objects i.e., the edges are brighter than the rest of the image. When we enhance the edges by adding the output of an edge detection filter (like Sobel) to the original image, both edge and non-edge pixels can be affected depending on scaling and normalization. The edge pixels become brighter whereas the intensity of the non-edge pixels reduces.

6 CONCLUSIONS

Sobel filters estimate the first-order derivative and detect edges based on gradient magnitude along x and y directions separately. These filters are less sensitive to noise and produce thicker and smoother edges. Laplacian filters, on the other hand, estimate the second-order derivative and detect edges based on zero-crossings in the second derivative. These are more sensitive to noise and produce thinner and sharper edges.

Sobel usually works better for edge enhancement in noisy or complex images because it combines edge detection and slight smoothing, producing edges that consist of less noise.

Laplacian can produce sharper edges but tends to amplify noise, so it is often used with Gaussian smoothing beforehand, or combined with other filters to get cleaner results. If noise is low and very crisp edges are required, Laplacian can be advantageous. For general enhancement, Sobel is preferred due to robustness.

7 REFERENCES:

1. X. Ren and S. Lai, "Medical Image Enhancement Based on Laplace Transform, Sobel Operator and Histogram Equalization," Academic Journal of Computing & Information Science, vol. 5, no. 6, pp. 48-54, 2022, Doi: 10.25236/AJCIS.2022.050608.
2. L. Haidar, "Sobel vs. Canny Edge Detection Techniques: Step by Step Implementation," Medium, Sep. 11, 2021. [Online]. Available: https://medium.com/@haidarlina4/sobel-vs-canny-edge-detection-techniques-step-by-step-implementation-11ae6103a56a.
3. H. K. Joy, "Understanding First Order Derivative Methods in Edge Detection," Medium, Oct. 20, 2024. [Online]. Available: https://medium.com/@helenjoy88/understanding-first-order-derivative-methods-in-edge-detection-0d5d81d25a78.